Chicago Traffic Crashes

COP 3045 - Python Programming II

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Team 3

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1. Introduction

This project analyzes traffic crashes in Chicago. The goal is to explore how weather, lighting, time of day, speed limits, and the number of vehicles are related to crashes and injuries.

The project uses two related datasets from the City of Chicago Data Portal. The datasets are connected using CRASH_RECORD_ID.

2. Data Sources

Dataset 1: Traffic Crashes - Crashes
Source: City of Chicago Data Portal
Dataset ID: 85ca-t3if

Dataset 2: Traffic Crashes - Vehicles
Source: City of Chicago Data Portal
Dataset ID: 68nd-jvt3

The vehicle dataset was grouped by crash ID to calculate the number of vehicles in each crash. The counts were then merged with the crash dataset.

3. Dataset Summary

Time period: March 03, 2013 to October 10, 2026

Total crash records: 1,100,071
Total recorded injuries: 221,199
Average injuries per crash: 0.20
Average vehicles per crash: 2.04

The data includes numerical variables such as injuries, speed limit, and number of vehicles. Categorical variables include weather condition, lighting condition, and crash type.

4. Exploratory Data Analysis

The following tables summarize traffic crash patterns across different conditions. Only the first 10 selected rows of each result are shown.

Weather Conditions

Weather Condition	Total Crashes	Average Injuries	Total Injuries
CLEAR	863,465	0.21	178,940
RAIN	91,215	0.25	22,505
UNKNOWN	68,499	0.07	4,619
SNOW	35,021	0.17	5,913
CLOUDY/OVERCAST	32,150	0.21	6,838
OTHER	3,498	0.25	883
FREEZING RAIN/DRIZZLE	2,571	0.23	601
FOG/SMOKE/HAZE	1,621	0.25	413
SLEET/HAIL	1,225	0.23	287
BLOWING SNOW	624	0.27	166

Lighting Conditions

Lighting Condition	Total Crashes	Average Injuries	Total Injuries
DAYLIGHT	707,384	0.19	134,879
DARKNESS, LIGHTED ROAD	237,549	0.27	62,956
UNKNOWN	55,665	0.05	3,019
DARKNESS	50,594	0.19	9,614
DUSK	30,736	0.22	6,637
DAWN	18,143	0.23	4,094

Crashes by Hour

Crash Hour	Total Crashes	Total Injuries
15	85,834	15,656
16	84,409	15,983
17	81,737	15,873
14	73,414	13,595
18	67,052	13,715
13	66,305	12,090
12	64,011	11,016
8	59,132	10,621
11	55,616	9,821
9	50,348	8,678

Posted Speed Limits

Posted Speed Limit	Total Crashes	Average Injuries
30	812,702	0.21
25	72,891	0.16
35	68,942	0.27
20	46,597	0.12
15	38,892	0.08
10	26,532	0.05
40	10,623	0.32
0	7,891	0.09
45	7,080	0.30
5	5,589	0.07

Crash Severity

Crash Type	Total Crashes	Average Injuries	Total Injuries
NO INJURY / DRIVE AWAY	801,551	0.00	25
INJURY AND / OR TOW DU...	298,520	0.74	221,174

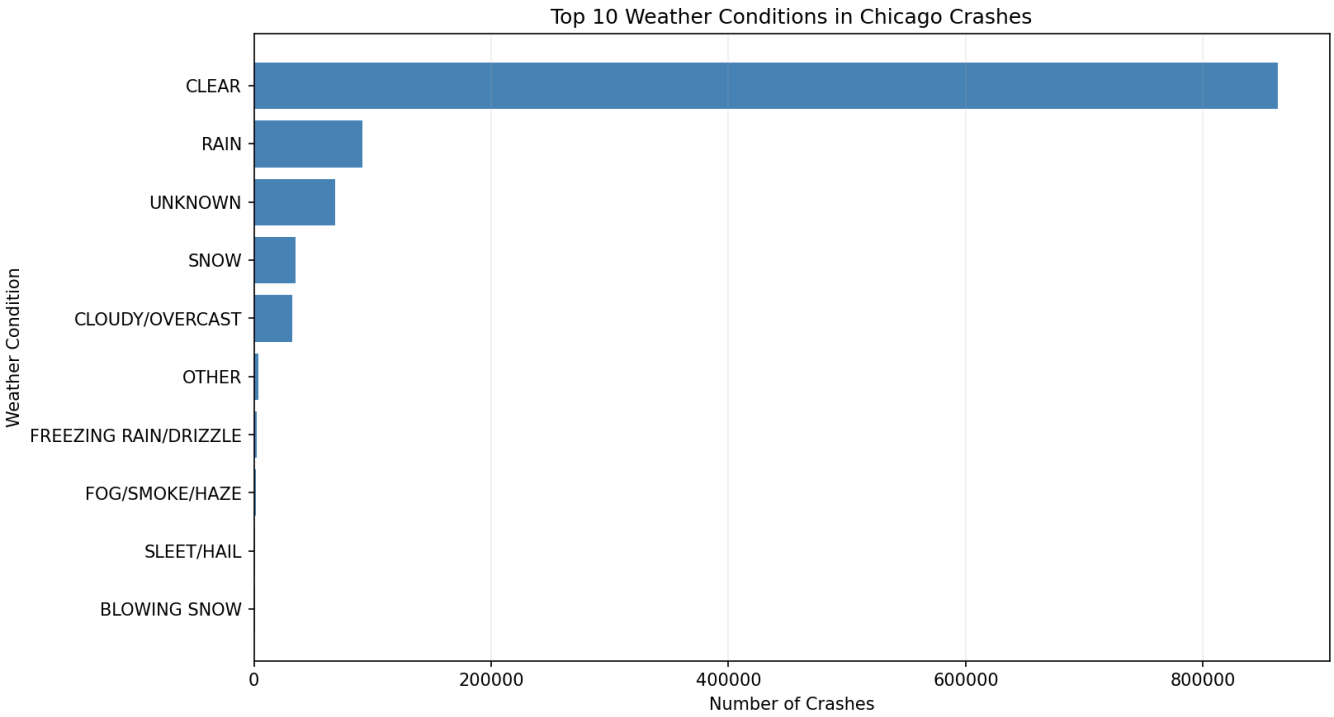
Crashes by Month

Crash Month	Total Crashes
1	83597
2	80724
3	85046
4	84125
5	97199
6	97410
7	97997
8	100343
9	100714
10	98331

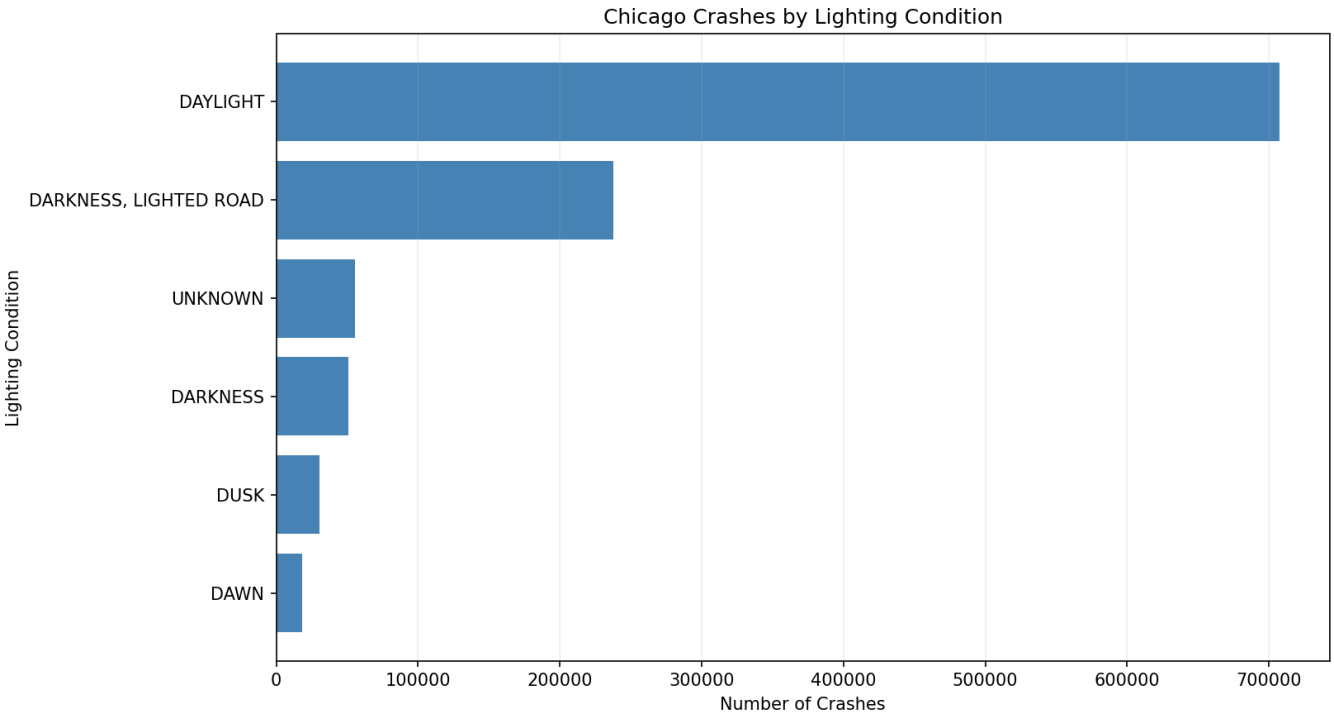
Number of Vehicles

Num Vehicles	Total Crashes	Average Injuries	Total Injuries
1	58,077	0.21	12,221
2	964,750	0.18	171,332
3	59,856	0.45	26,982
4	13,042	0.57	7,426
5	2,852	0.71	2,027
6	953	0.76	728
7	268	0.91	245
8	142	0.75	107
9	56	1.04	58
10	35	0.83	29

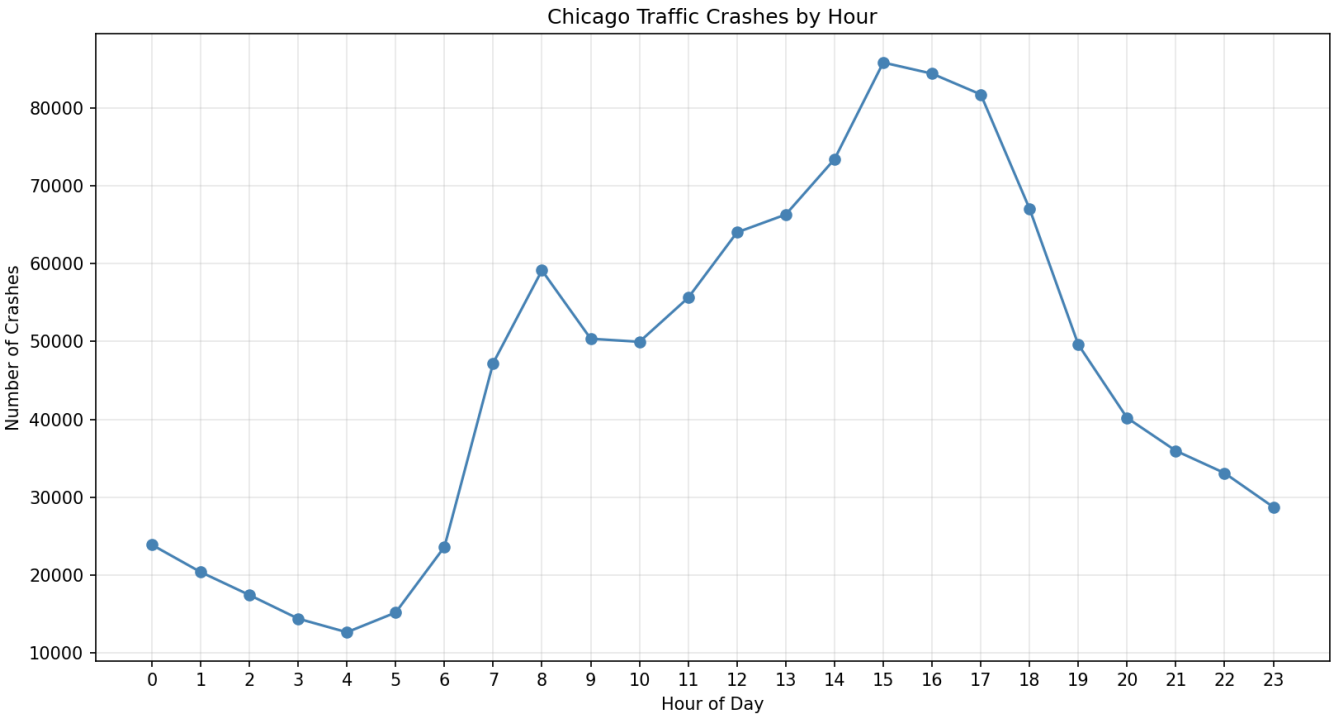
5.1 Crashes by Weather



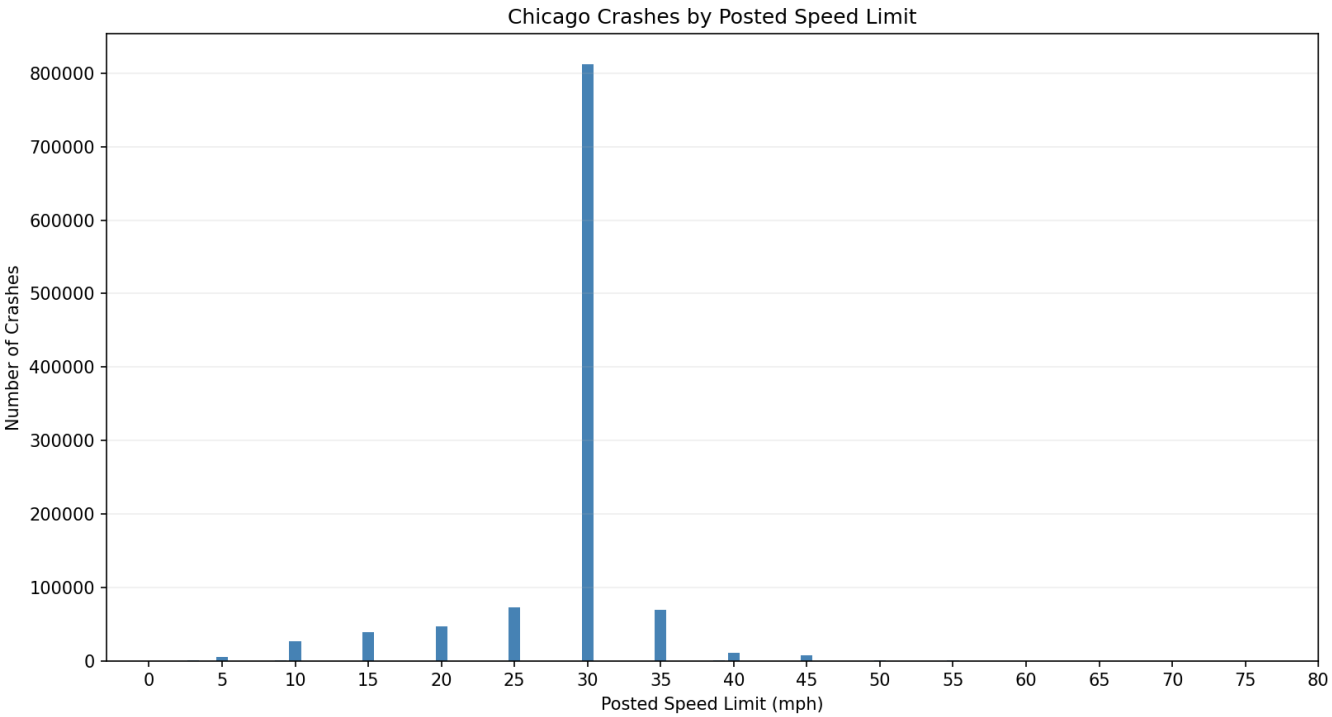
5.2 Crashes by Lighting



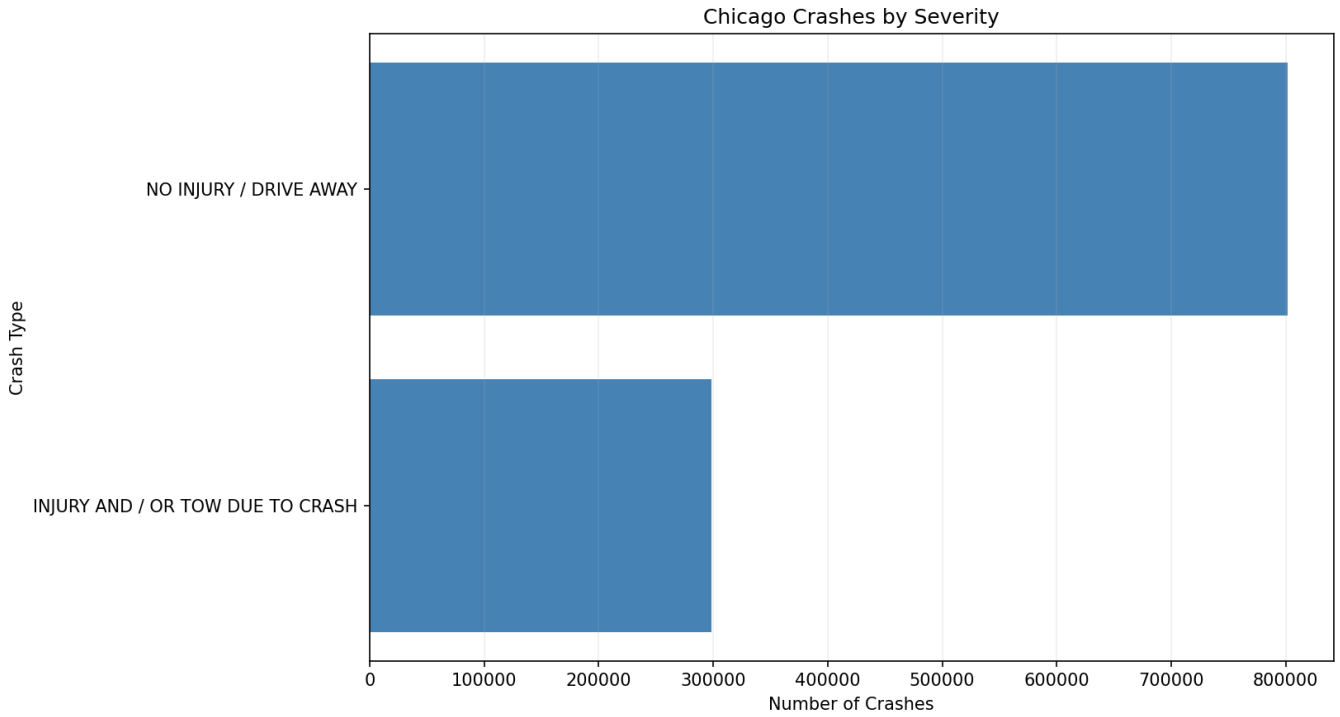
5.3 Crashes by Hour



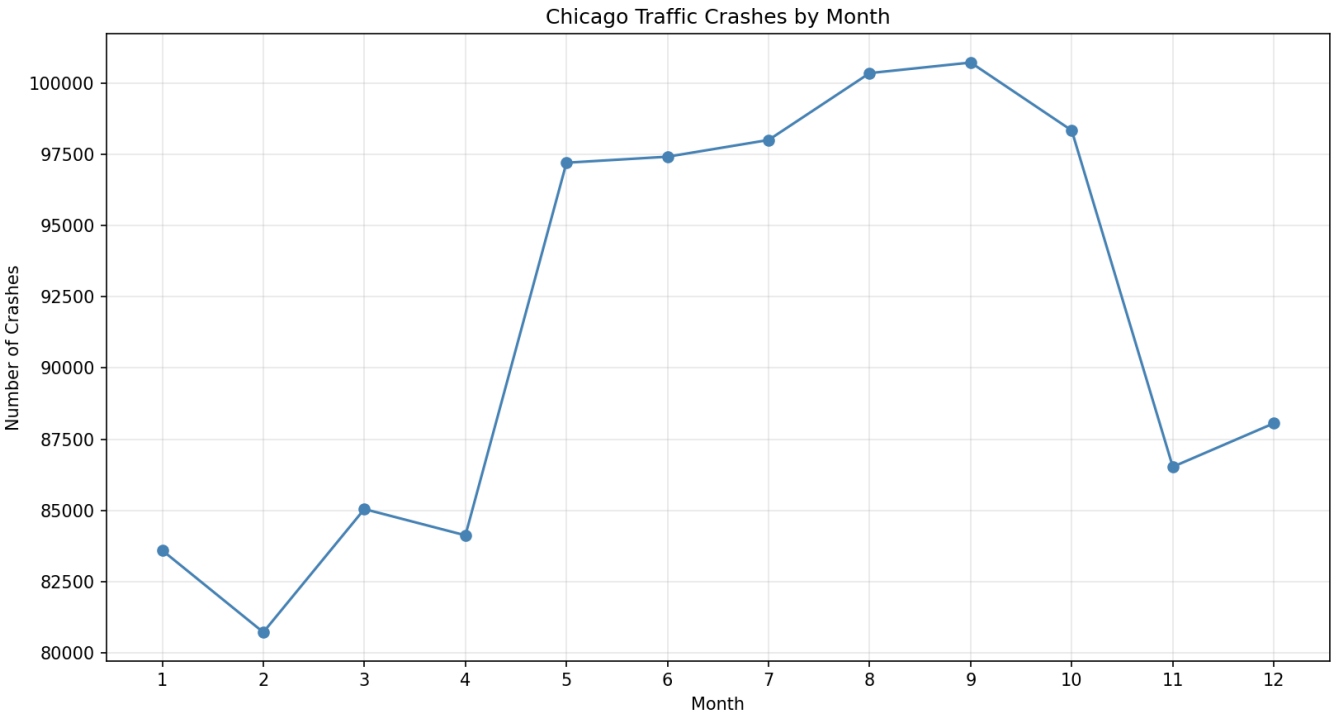
5.4 Crashes by Speed Limit



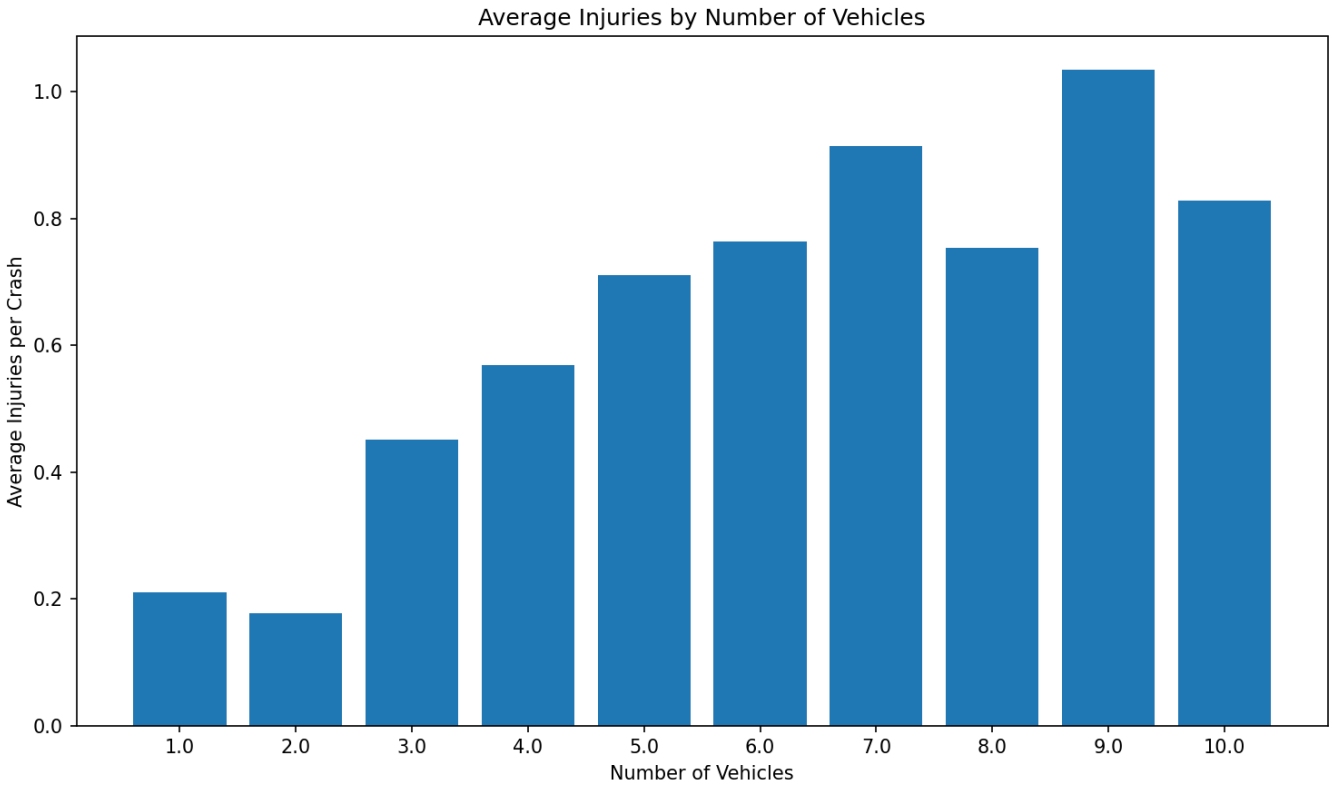
5.5 Crash Severity



5.6 Crashes by Month



5.7 Injuries by Number of Vehicles



6. Key Findings

The most common recorded weather condition was CLEAR, with 863,465 crashes.

The most common recorded lighting condition was DAYLIGHT, with 707,384 crashes.

The hour with the highest recorded number of crashes was 15:00, with 85,834 crashes.

The most common number of vehicles involved in a crash was 2, with 964,750 recorded crashes.

The analysis also compares average injuries across crashes involving different numbers of vehicles.

These findings describe patterns in the dataset. They do not prove that any specific condition causes a crash.

7. Conclusion

This project demonstrates an automated data analysis workflow using Python.

The application reads and prepares two related datasets, creates summary statistics and charts, and automatically generates a PDF report.

Pandas was used for data processing and analysis, Matplotlib for creating charts, and fpdf2 for generating the report.